GIANLUIGI VITALE

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RESEARCH INTERESTS

Machine learning systems, LLM serving infrastructure, accelerator portability, TPU kernels and numerical behavior, ML benchmarking, inference reliability, and LLM safety evaluation.

EDUCATION

B.Sc. in Computer Engineering — Universitas Mercatorum, Rome, Italy *Aug. 2024–Mar. 2027*

GPA: 29.35/30 (Italian scale) · 3.87/4.0 (WES-converted) · expected final grade 110/110 cum laude — all coursework completed; thesis in progress (graduation expected Mar. 2027)

Thesis: Natural Language Is Flatland for Machine Intelligence: AI Systems Should Decouple Their Reasoning Substrate from Human Language

Supervisor: Prof. Rocco Pietrini

Full Stack Web Development Bootcamp — *Boolean SRL (700 hours)* *Jan.–Jun. 2020*

High-school diploma (Diploma di maturità) *2019*

PUBLICATIONS

G. Vitale, "Decomposing One Professional-Framing Pipeline: Which Components Shift LLM Safety Boundaries?" **NeurIPS 2026**, main track (poster). Accepted; to appear.
Paper | Poster

G. Vitale, "DriftBench: Measuring and Predicting Infrastructure Drift in LLM Serving Systems," **MLSys 2026**, Bellevue, WA, May 2026. Oral presentation.
Paper | Artifact (ACM badges: Available, Functional, Reproduced) | Talk

MANUSCRIPTS UNDER REVIEW

G. Vitale, R. Pietrini, "Cross-Generation Numerical Agreement of LLM Inference on TPUs," under review at **MLSys 2027**.

HONORS & AWARDS

Distinguished Artifact Evaluator Award — *SOSP 2026 Artifact Evaluation Committee* *2026*

RESEARCH EXPERIENCE

- **DriftBench and TPU validation:** Built a measurement and prediction framework spanning 236,985 prompt-response pairs across 105 configurations. The Portability Risk Index generalizes to unseen hardware ($R^2 = 0.909$). Extended the result to TPU v5e/v6e, where predictability held across a new numerical path ($R^2 = 0.72$–0.96).
- **Factorial ablation of jailbreak framing (NeurIPS 2026):** Applied a pre-specified 2×2×2 factorial design to the STF professional-framing pipeline: 5,000 human-labeled trials on three closed frontier models in the primary study, four 5,000-trial factorials (20,000 trials) in total, and behavioral checks on six open-weight models. A professional persona was the main driver of acceptance (OR = 6.5), Linguistic Substitution the main driver of response depth (cumulative-logit coefficient $\beta = 0.91$), and moral justification showed no detectable positive effect.
- **Bit-level TPU MXU characterization (co-authored with Prof. Rocco Pietrini; under review at MLSys 2027):** Probing TPU v4, v5e, and v6e from outside, we localized cross-generation bf16 matmul differences to the compiled cross-pass reduction order. A CPU emulator reproduced 34.6M fresh outputs bit for bit on all three generations (bf16, default precision, four standard input regimes). Matching matmuls alone does not make the model agree; whole-program pinning yields identical logits and greedy text on all three generations, and identical vLLM output on v5e/v6e.
- **DeepSeek V4-Flash on TPU:** Completed a 32-chip v4-64 bring-up, validating every attention mode with both prefill and decode operational. Equivalent support has since landed in vLLM TPU inference through parallel development in the wider ecosystem.

- **DSA kernel for TPU:** Developed a working implementation that is bit-exact against the reference through 8K tokens and runs through 256K. Work in progress; it underpins the GLM-5.3 port, now running on TPU v4.
- **Long-context genomics with ICAR-CNR:** Collaborating on whether long-context DNA models encode linkage disequilibrium, combining ML systems work with genomics expertise at ICAR-CNR (Italy's National Research Council).

ACADEMIC SERVICE

Program Committee Member — *MLSys 2027* *2026–2027*

Reviewing submissions for the MLSys 2027 technical program.

Ethics Reviewer — *NeurIPS 2026* *2026*

Reviewed submissions for ethical concerns, potential societal harms, research-integrity risks, and compliance with the conference's ethics guidelines.

Artifact Evaluation Committee Member — *SOSP 2026* *2026*

Evaluated research artifacts in memory-efficient model serving and memory-safety analysis for GPU kernels in LLM inference systems. Assessed availability, functionality, reproducibility, and consistency with the associated papers' experimental claims.

Artifact Evaluation Committee Member — *MLSys 2026* *2026*

Evaluated four accepted submissions spanning attention-kernel optimization, ML profiling infrastructure, GPU memory disaggregation, and distributed inference. Reproduced experimental results on rented 8×H100 clusters (RunPod) and AMD MI350X infrastructure (AMD University Program), assessing availability, functionality, and consistency with each paper's claims.

PROFESSIONAL EXPERIENCE

Systems Engineer — *State Archive of Pistoia, Italian Ministry of Culture* *Apr. 2024–Present*

Designed and deployed a retrieval-augmented generation system over 10,000 internal administrative documents, delivering source-cited answers to staff queries. Built an AI-assisted public consultation interface giving staff and citizens natural-language access to the full inventory database.

Selection Preparation and Independent Study *Apr. 2021–Mar. 2024*

Prepared for the selection of a specialized cybercrime unit of Italy's financial police (Guardia di Finanza — Nucleo Speciale Tutela Privacy e Frodi Tecnologiche): completed the required military preparation (physical training: strength, agility, endurance) and passed all technical and physical examinations. In parallel, studied computer science independently, then won the public competition for the Systems Engineer post (Apr. 2024).

Web Developer — *Informamuse S.R.L.* *Oct. 2020–Apr. 2021*

Developed websites and web applications using HTML, CSS, JavaScript, PHP, Laravel, Vue.js, and MySQL. Managed projects end-to-end from design through deployment.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, HTML/CSS, Bash

Frameworks: PyTorch, JAX, Pallas, Hugging Face Transformers

ML/Systems: vLLM, tpu-inference, SGLang, TensorRT-LLM, CUDA, Docker

Infrastructure: Google Cloud TPU (v4, v5e, v6e), Linux administration, networking, cloud GPU provisioning (RunPod, AMD University Program, Chameleon Cloud), CI/CD

Research Tools: LaTeX, Git, Jupyter, scipy, statsmodels

COMPUTATIONAL RESOURCES

- Google Cloud TPU Research Cloud (TRC) — active since May 2026. Current 320-chip allocation: 32 on-demand v4 plus 288 spot chips across v4, v5e, and v6e in US and EU zones.
- AMD University Program account (active since March 2026); Chameleon Cloud allocation (2,000 SUs); RunPod GPU cluster experience (H100, H200, B200, MI300X)

LANGUAGES

Italian (native), **English** (C1 — academic proficiency, scientific publications)